CAMPAIGN RESULTS

# Mixed Training Campaign Results Report

Mixed Feedforward Campaign Closing Report

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

## Overview

This report summarizes the completed mixed feedforward training campaign prepared in:

- `doc/reports/campaign_plans/2026-03-12-15-32-28_mixed_training_campaign_plan_report.md`

The campaign was designed to test whether the next meaningful improvement over the earlier feedforward baseline family would come from:

- denser point sampling;
- larger effective batch sizes;
- larger model capacity;
- or interactions between those levers on top of the strongest long training schedule.

The campaign executed successfully end to end:

- completed runs: `9`
- failed runs: `0`
- queue status after completion: no pending, running, or failed campaign items

The execution artifact root is:

- `output/training_campaigns/2026-03-12-18-52-30_mixed_training_campaign_2026_03_12_15_32_28/`

## Historical Reference Set

Before this mixed campaign, the feedforward project already had five important reference configurations:

- `trial`
- `baseline`
- `high_density`
- `high_epoch`
- `high_compute`

These runs remain useful as context because they define the baseline tradeoffs that motivated the mixed campaign.

| Config | Status | Best Epoch | Approx. Wall Time | Val MAE [deg] | Val RMSE [deg] | Test MAE [deg] | Test RMSE [deg] |
|---|---|---|---|---|---|---|---|
| `trial` | historical reference | 3 | ~5.0 min | 0.003757 | 0.004555 | 0.003583 | 0.004295 |
| `baseline` | historical reference | 22 | ~11.2 min | 0.003148 | 0.003655 | 0.003504 | 0.003969 |
| `high_density` | historical reference | 41 | ~12.6 min | 0.003077 | 0.003642 | 0.003519 | 0.004046 |
| `high_epoch` | historical reference | 94 | ~24.2 min | 0.003007 | 0.003451 | 0.003335 | 0.003767 |
| `high_compute` | historical reference | 21 | ~10.6 min | 0.003198 | 0.003790 | 0.003319 | 0.003915 |

The earlier campaign already showed one clear result:

- longer optimization helped more than denser point sampling alone.

  That is why the new campaign kept the long schedule and explored density, batch, and capacity combinations around it.

## Mixed Campaign Setup

The `9` new runs were organized in three phases:

- Phase 1

  Density plus long schedule.
- Phase 2

  Density plus long schedule plus larger batch.
- Phase 3

  Density plus long schedule plus larger batch plus larger model. All runs kept the same:
- dataset split logic;
- feature set ( `input_size = 5` );
- target definition;
- optimizer family ( `AdamW` );
- held-out validation and test workflow.

## Phase 1 Results

Phase 1 asked a focused question

- if we keep the strongest long schedule, does denser sampling alone become enough to beat the historical `high_epoch` reference?

| Config | Best Epoch | Wall Time | Val MAE [deg] | Val RMSE [deg] | Test MAE [deg] | Test RMSE [deg] |
|---|---|---|---|---|---|---|
| `stride10_long` | 107 | 23.8 min | 0.003053 | 0.003588 | 0.003483 | 0.004050 |
| `stride5_long` | 43 | 17.2 min | 0.003178 | 0.003660 | 0.003580 | 0.004008 |
| `stride1_long` | 66 | 37.7 min | 0.003126 | 0.003514 | 0.003646 | 0.003990 |

### Interpretation

Phase 1 did not overturn the earlier conclusion.

- `stride10_long` was the best Phase 1 candidate, but it still stayed behind `high_epoch` on held-out test metrics.
- `stride5_long` did not justify its stronger density setting.
- `stride1_long` was the most expensive Phase 1 run and also the weakest on `test_mae` .

  The practical message is still the same:
- denser sampling alone is not enough;
- especially full-density training is not automatically worth the cost.

## Phase 2 Results

Phase 2 asked the next question

- if density is paired with a larger batch regime, do the denser variants become more competitive?

| Config | Best Epoch | Wall Time | Val MAE [deg] | Val RMSE [deg] | Test MAE [deg] | Test RMSE [deg] |
|---|---|---|---|---|---|---|
| `stride10_long_large_batch` | 111 | 18.4 min | 0.003066 | 0.003765 | 0.003433 | 0.004123 |
| `stride5_long_large_batch` | 81 | 16.8 min | 0.003109 | 0.003639 | 0.003301 | 0.003791 |
| `stride1_long_large_batch` | 50 | 29.2 min | 0.003104 | 0.003551 | 0.003358 | 0.003769 |

### Interpretation

Phase 2 is the strongest part of the whole campaign.

- `stride5_long_large_batch` achieved the best `test_mae` not only within the mixed campaign, but across the full feedforward set currently available in the repository.
- `stride1_long_large_batch` produced a very strong `test_rmse`, nearly tied with the historical `high_epoch` reference.
- `stride10_long_large_batch` did not benefit enough from the larger batch to justify becoming the new default.

The important takeaway is:

- a larger batch helped when paired with the denser settings;
- the best practical gain came from the middle density regime ( `stride = 5` ), not from the full-density extreme.

## Phase 3 Results

Phase 3 asked the final question

- after strengthening density and batch, does larger model capacity finally become worth it?

| Config | Best Epoch | Wall Time | Val MAE [deg] | Val RMSE [deg] | Test MAE [deg] | Test RMSE [deg] |
|---|---|---|---|---|---|---|
| `stride10_long_large_batch_big_model` | 67 | 19.8 min | 0.003040 | 0.003702 | 0.003413 | 0.004063 |
| `stride5_long_large_batch_big_model` | 104 | 33.8 min | 0.003104 | 0.003666 | 0.003472 | 0.004004 |
| `stride1_long_large_batch_big_model` | 78 | 56.7 min | 0.003090 | 0.003585 | 0.003308 | 0.003779 |

### Interpretation

Phase 3 confirms that larger capacity is still not the main answer.

- the best large-model variant was `stride1_long_large_batch_big_model`, but it did not produce a decisive gain over the simpler Phase 2 leaders.
- `stride5_long_large_batch_big_model` was slower and worse than `stride5_long_large_batch`.
- `stride1_long_large_batch_big_model` remained very competitive on `test_mae`, but at a much larger runtime cost.

The larger model therefore looks unnecessary as the default feedforward setting at this stage.

## Cross-Campaign Ranking

The completed campaign clarifies the top of the feedforward ranking.

### Best Test MAE

| Config | Test MAE [deg] | Test RMSE [deg] | Runtime |
|---|---|---|---|
| `stride5_long_large_batch` | 0.003301 | 0.003791 | 16.8 min |
| `stride1_long_large_batch_big_model` | 0.003308 | 0.003779 | 56.7 min |
| `high_compute` | 0.003319 | 0.003915 | 10.6 min |
| `high_epoch` | 0.003335 | 0.003767 | 24.2 min |
| `stride1_long_large_batch` | 0.003358 | 0.003769 | 29.2 min |

### Best Test RMSE

| Config | Test RMSE [deg] | Test MAE [deg] | Runtime |
|---|---|---|---|
| `high_epoch` | 0.003767 | 0.003335 | 24.2 min |
| `stride1_long_large_batch` | 0.003769 | 0.003358 | 29.2 min |
| `stride1_long_large_batch_big_model` | 0.003779 | 0.003308 | 56.7 min |
| `stride5_long_large_batch` | 0.003791 | 0.003301 | 16.8 min |
| `high_compute` | 0.003915 | 0.003319 | 10.6 min |

## Main Conclusions

The mixed campaign supports five practical conclusions.

### 1. The Long Schedule Remains Mandatory

Nothing in the new campaign argues for returning to the shorter baseline schedule. The long optimization regime remains the right backbone for serious feedforward training.

### 2. Full-Density Training Is Not The Best Default

`stride = 1` can be competitive, but it is expensive and not decisively superior. The full-density regime should remain a targeted benchmark, not the standard configuration.

### 3. The Middle-Density Regime Is The Best Practical Upgrade

`stride5_long_large_batch` is the most convincing practical winner:

- best global `test_mae`;
- much lower runtime than `high_epoch`;
- standard model capacity;
- no reliance on the heavier large-model architecture.

### 4. Larger Batch Helped More Than Larger Model Capacity

The strongest gains came from the Phase 2 batch increase, not from the Phase 3 model enlargement.

### 5. The Historical `high_epoch` Reference Still Matters

`high_epoch` remains slightly best on `test_rmse`. This means it is still a valuable secondary reference whenever the project wants the cleanest low-variance historical baseline.

## Recommended Best Training Preset

The new recommended practical feedforward preset should be:

- `config/training/feedforward/presets/best_training.yaml`

This preset should follow `stride5_long_large_batch`. Why this is the best current practical choice:

- it achieved the best held-out `test_mae` across the current feedforward configurations;
- it kept the standard `128-128-64` architecture;
- it avoided the extra cost and complexity of the larger model;
- it finished faster than `high_epoch` while still improving the main absolute-error metric.

Important nuance:

- if the project later decides that `test_rmse` must dominate the ranking more strongly than `test_mae`, `high_epoch` remains the cleanest alternative reference.

## Recommended Next Experiments

The next useful feedforward experiments are

1. repeat `best_training` with multiple random seeds;
2. compare `best_training` and `high_epoch` across multiple seeds to separate real gain from split-level variance;
3. perform regime-wise analysis by speed, torque, and temperature;
4. test whether a slightly lower learning rate further improves the `stride5_long_large_batch` convergence profile;
5. keep the large-model variants as optional offline benchmarks rather than default training presets.

## Artifact References

Campaign-level references

- `output/training_campaigns/2026-03-12-18-52-30_mixed_training_campaign_2026_03_12_15_32_28/campaign_manifest.yaml`
- `output/training_campaigns/2026-03-12-18-52-30_mixed_training_campaign_2026_03_12_15_32_28/campaign_execution_report.md`

Best practical mixed-campaign run:

- `output/feedforward_network/te_feedforward_stride5_long_large_batch/`

Strong historical comparison run:

- `output/feedforward_network/te_feedforward_high_epoch/`

Each run folder includes:

- `best_checkpoint_path.txt`
- `training_test_metrics.yaml`
- `training_test_report.md`
- checkpoint files